

Milestone 4: Final Review and Delivery (Weeks 7–8)

1. Introduction

Milestone 4 represents the final stage of the Unified Military Analytics and Comparison Dashboard project. After completing data collection, KPI engineering, and dashboard development in the previous milestones, this phase focuses on validation, refinement, and final delivery.

The goal of this milestone is to ensure that the entire analytical system is reliable, accurate, user-friendly, and ready for academic submission. While previous milestones emphasized development and implementation, this stage emphasizes verification and presentation quality.

This milestone ensures that the dashboard operates as a complete analytical solution rather than a prototype.

Milestone 4 consists of two modules:

- Module 7: Testing and Debugging
- Module 8: Documentation and GitHub Release

These modules ensure that the project meets academic requirements and can be reproduced by other users.

2. Module 7: Testing and Debugging

2.1 Objective

The objective of Module 7 is to verify the correctness and stability of the dashboard system. Interactive dashboards involve multiple filters, navigation links, and calculated measures, which must be tested thoroughly to avoid incorrect results.

Testing ensures that:

- All calculations produce correct values
- Filters behave as expected
- Navigation works properly
- Visualizations update dynamically
- No errors appear during interaction

This module transforms the dashboard from a development product into a reliable analytical tool.

2.2 Filter and Slicer Testing

Extensive testing was conducted on all filters and slicers across dashboards.

Country Filters

Country selection was tested to verify that:

- Nation Overview displays correct country data
- Compare Powers updates correctly
- Coalition Builder aggregates values accurately
- Multiple countries were selected and removed to confirm dynamic updates.

Region Filters

Region filters were tested to confirm:

- Countries are correctly grouped by region
- Visualizations update immediately after selection
- Coalition calculations change accordingly

Region filtering enables realistic coalition simulation and therefore required careful verification.

NATO Filters

Alliance filters were tested to ensure:

- NATO and Non-NATO countries are correctly classified
- NATO contribution charts update correctly
- Coalition NATO shares reflect selected countries

The NATO classification was verified using the mapping implemented in Python.

2.3 KPI Validation

All KPI values displayed in dashboards were verified through manual spot-checking against the source dataset.

Verification steps included:

- Comparing defense budget totals with dataset values
- Checking manpower aggregation for coalitions
- Confirming assets-per-capita calculations
- Validating composite power scores

Random countries were selected and values were manually confirmed using the dataset.

This process ensured that no calculation errors occurred during aggregation.

2.4 Aggregation Logic Verification

Different metrics require different aggregation methods. Each aggregation was verified to ensure correctness.

SUM Aggregations

Used for:

- Defense Budget
- Active Personnel
- Military Assets

These metrics represent totals and therefore must be summed.

AVERAGE Aggregations

Used for:

- Composite Power Score
- Assets per Capita
- Budget Ratios

These metrics represent normalized indicators and therefore must be averaged.

2.5 Dashboard Interaction Testing

Each dashboard page was tested to verify interactive behavior.

Quick Stats Dashboard Testing

Verified that:

- Top 10 ranking updates dynamically
- KPI cards change with filters
- Charts respond to region selections
- No blank visuals appear

Nation Overview Dashboard Testing

Verified that:

- Country selection updates all visuals
- Asset composition displays correctly
- Personnel structure updates properly
- NATO status changes correctly
- Map displays correct country location

Compare Powers Dashboard Testing

Verified that:

- Country selection updates comparison charts
- Both countries display correctly
- Budget comparisons update properly
- Personnel comparisons update correctly

Multiple country combinations were tested to ensure consistency.

Coalition Builder Dashboard Testing

Coalition Builder required the most extensive testing because of dynamic aggregation.

Testing confirmed that:

- Coalition totals update correctly
- Adding countries increases totals
- Removing countries decreases totals
- NATO contributions update correctly
- Regional coalitions work properly

Coalition scenarios such as regional alliances and NATO alliances were tested.

2.6 Navigation Testing

Navigation buttons were tested across all dashboards.

Testing verified:

- Correct page navigation
- No broken links
- Smooth transitions

Navigation was tested in both:

- Edit mode (Ctrl + Click)
- Reading mode

This ensures the dashboard works correctly during presentation and evaluation.

2.7 Layout and Visual Refinements

Several refinements were made to improve visual quality and usability.

Formatting Improvements

Changes included:

- Standardized number formatting
- Removal of scientific notation
- Improved chart titles
- Clear labeling of KPIs

These improvements enhance readability and professionalism.

Alignment Improvements

Visual alignment was standardized across pages to ensure consistent layout.

Uniform spacing was applied between:

- KPI cards
- Charts
- Navigation buttons

This improves visual balance.

Color Standardization

A consistent color scheme was applied across dashboards.

Consistent colors help users identify:

- Budget metrics
- Asset metrics
- Personnel metrics
- NATO contributions

This improves interpretability.

2.8 Performance Testing

Performance testing was conducted to ensure smooth dashboard operation.

Tests confirmed:

- Fast filter response
- Stable navigation
- No visual loading issues
- Smooth coalition aggregation

The dashboard performs efficiently even when multiple countries are selected.

2.9 Quality Assurance Checklist

A final quality assurance checklist was used.

The following items were verified:

- All KPIs display correct values
- Filters update visuals correctly
- Navigation buttons function properly
- Charts display correct labels
- No empty visuals appear
- Coalition calculations are correct
- Region filtering works correctly
- NATO classification is accurate

This checklist ensured readiness for final submission.

3. Module 8: Documentation and GitHub Release

3.1 Objective

The objective of Module 8 is to prepare the project for submission and reproducibility.

Proper documentation ensures that:

- The methodology is clear
- Scripts can be reused
- Data preparation is reproducible
- Dashboards can be understood easily

3.2 Project Documentation

Documentation was prepared to describe the entire workflow.

Documentation includes:

- Data scraping methodology
- Data cleaning steps
- KPI definitions
- Dashboard design
- Usage instructions

This documentation ensures transparency.

3.3 GitHub Repository Organization

The project repository was organized into structured directories.

Example structure:

/scripts

/data

/dashboard

/docs

Scripts Folder

Contains:

- scrape_military_metrics.py
- clean_data.ipynb
- generate_kpis.py

These scripts allow full reproduction of the dataset.

Data Folder

Contains:

- military_raw_data.csv
- military_cleaned.csv
- military_final.xlsx

These represent each stage of processing.

Dashboard Folder

Contains:

- Power BI dashboard file
- This is the final analytical product.

Documentation Folder

Contains:

- Milestone reports
- Project description

3.4 GitHub Version Control

The final version of the project was uploaded to GitHub.

Version control ensures:

- Traceability of updates
- Backup of project files
- Easy sharing

3.5 Dashboard Distribution

The dashboard can be opened in Power BI Desktop for interactive exploration.

Users can:

- Select countries
- Compare powers
- Build coalitions
- Explore global metrics

4. Milestone 4 Summary

Milestone 4 completes the Unified Military Analytics project by ensuring that the system is fully tested, polished, and documented.

This milestone delivers:

- Fully tested dashboards
- Verified calculations
- Clean documentation
- Organized repository

At the end of Milestone 4, the project represents a complete end-to-end data analytics system.

5. Final Project Completion

The project demonstrates the full data analytics lifecycle:

- Data scraping
- Data cleaning
- KPI engineering
- Dashboard development
- Testing and validation
- Documentation and release

The Unified Military Analytics Dashboard is now ready for academic submission and portfolio use.